Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**6**
Electrons (Ne)	**6**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**5**
Eigenstate idx	**4 (1-indexed: 5)**

Eigenstate properties

Energy E0	**-2.42291126**
Energy E1	**-2.09275383**
Gap (E1-E0)	**0.330157**
<S^2>	**2.000000**
S	**1.0000**
<Sz>	**0.000000**
Multiplicity	**3.0**
Spin label	**Triplet (S=1)**

Double occupancy

<D> total	**0.471901**
<D> / site	**0.078650**

RDM and GME statistics for this run (5-body clusters)

Hilbert space dim : 2^5 = 32
Number of clusters : 6
Pure clusters : 0
Mixed clusters : 6
Clusters with GME (>0) : 6

Purity gamma = Tr(rho^2) : mean = 0.399776 [min 0.399472 , max 0.400368]
Von Neumann entropy : mean = 1.156457 [min 1.153663 , max 1.158318]
Rank of rho : mean = 12.00 [min 12 , max 12]
C_GME (best estimate) : mean = 0.616118 [min 0.595199 , max 0.644032]

NOTE Analysis performed on eigenstate index 4 (state 5 in 1-indexed convention), not the ground state.

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Summary: All 5-Body RDM Clusters

System: L=6, Ne=6, U=4.0, t=1.0 | E_0=-2.422911

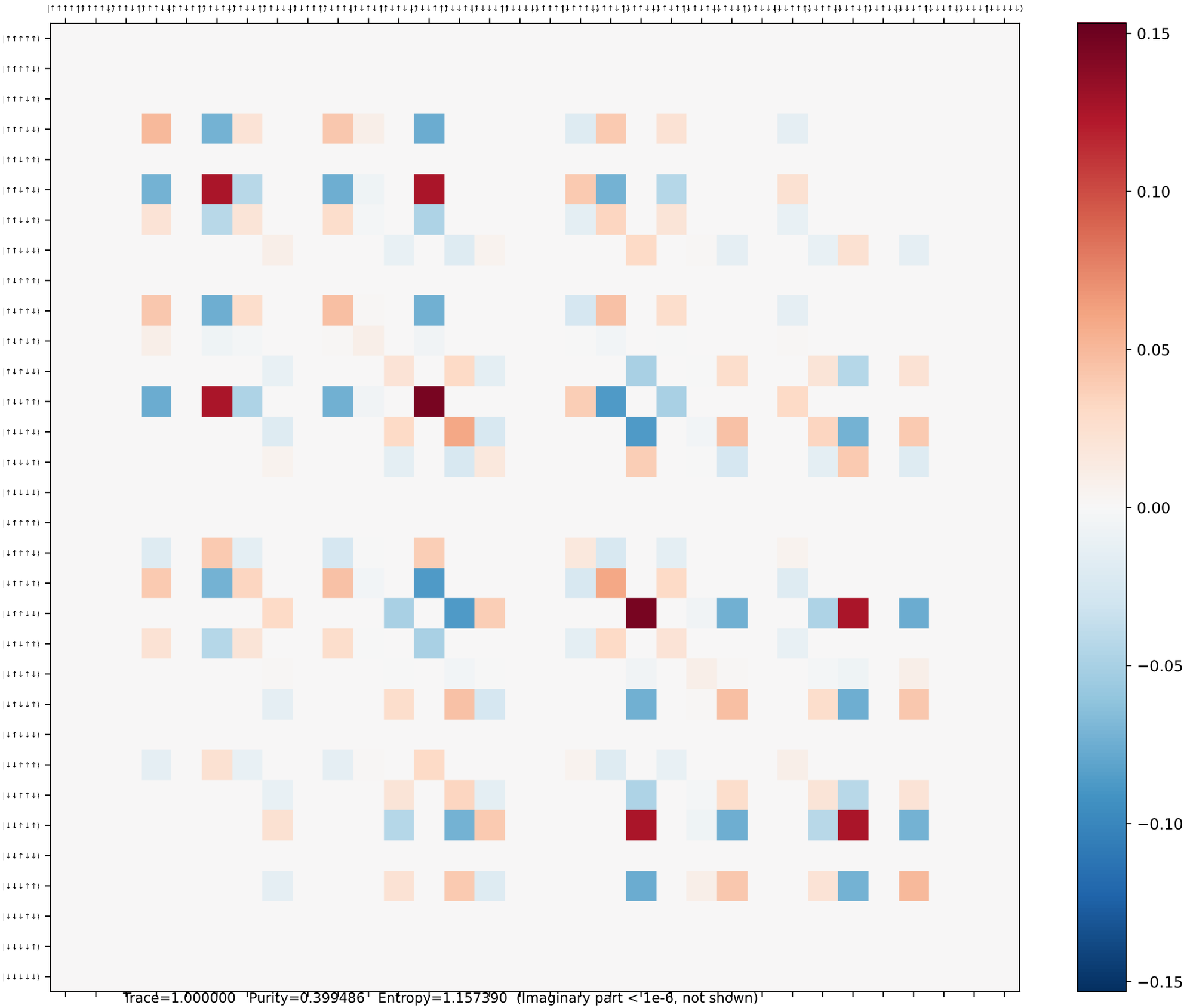
Sites	Purity	C_GME	C_LB	C_cayl	C_expo	C_stiefel	C_UB	Status
(0, 1, 2, 3, 4)	0.399486	–	0.000000	0.000000	0.000000	0.612063	0.791282	MIXED
(1, 2, 3, 4, 5)	0.400368	–	0.000000	0.000000	0.000000	0.644032	0.926891	MIXED
(2, 3, 4, 5, 0)	0.399472	–	0.000000	0.000000	0.000000	0.596040	0.790735	MIXED
(3, 4, 5, 0, 1)	0.399486	–	0.000000	0.000000	0.000000	0.612907	0.791299	MIXED
(4, 5, 0, 1, 2)	0.400368	–	0.000000	0.000000	0.000000	0.636468	0.926909	MIXED
(5, 0, 1, 2, 3)	0.399472	–	0.000000	0.000000	0.000000	0.595199	0.790902	MIXED

Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map | BEST
  C_UB = Upper bound (eigendecomp) | Status: GME =has GME | NO GME=separable | MIXED=mixed state

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

5-Body RDM (Sites 0,1,2,3,4)

Real Part



Trace=1.000000 Purity=0.399486 Entropy=1.157390 (Imaginary part <1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 5-Body RDM (Sites 0,1,2,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3994863173
  Von Neumann entropy  : 1.1573895658
  Eigenvalue min       : -1.748166e-17
  Eigenvalue max       : 4.456381e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.39948632

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

STATUS: MIXED STATE
____________________________________

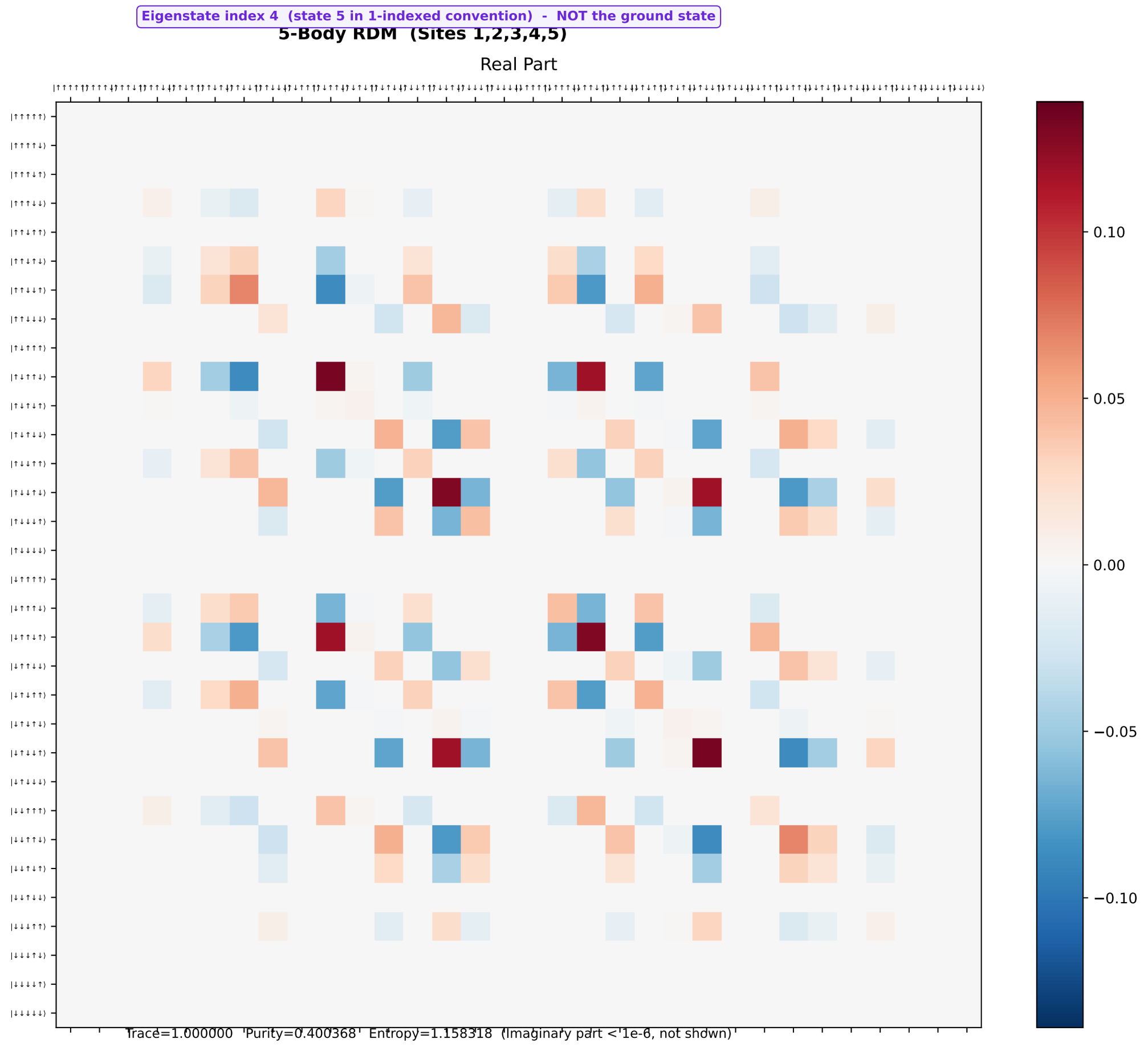
Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.61206285   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.79128167

  Status: nONE
```

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state



Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 5-Body RDM (Sites 1,2,3,4,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4003682228
  Von Neumann entropy  : 1.1583182995
  Eigenvalue min       : -4.023151e-17
  Eigenvalue max       : 4.462601e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.40036822

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

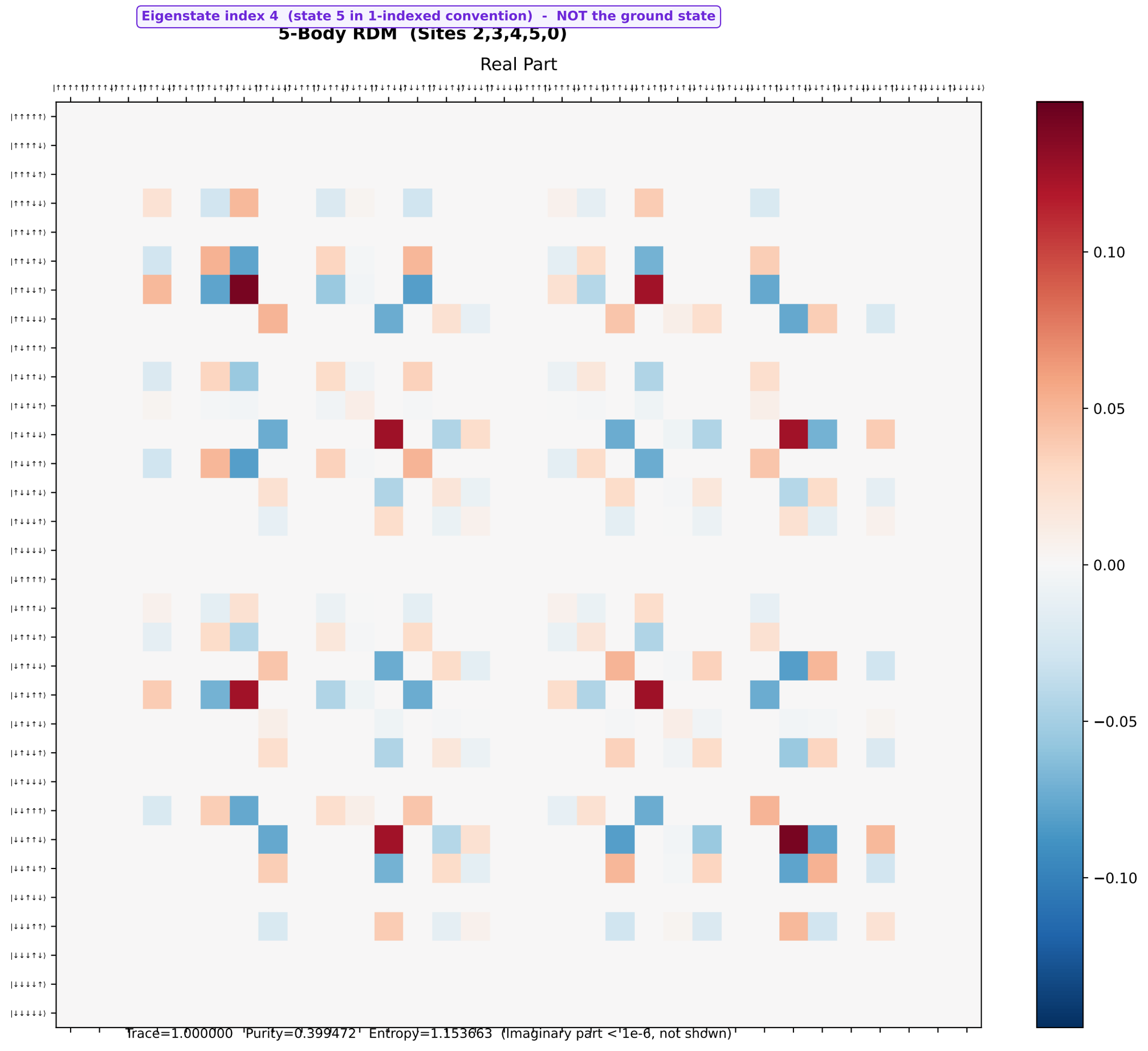
STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.64403190   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.92689078

  Status: nONE
```


Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state
5-Body RDM (Sites 2,3,4,5,0)
Real Part
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.399472 Entropy=1.153663 (Imaginary part <1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 5-Body RDM (Sites 2,3,4,5,0)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3994720399
  Von Neumann entropy  : 1.1536633312
  Eigenvalue min       : -8.510653e-17
  Eigenvalue max       : 4.455320e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.39947204

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

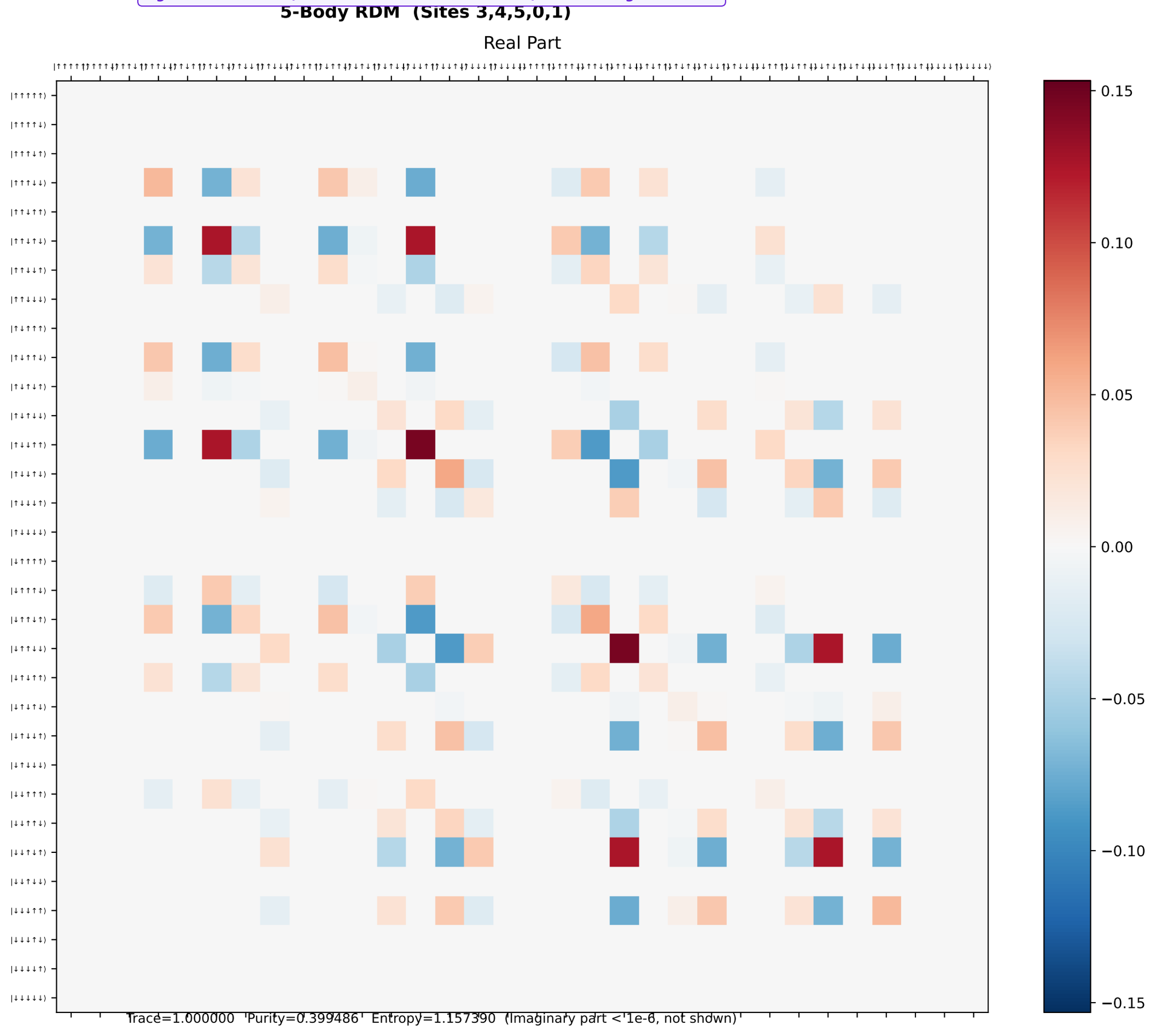
Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.59603955   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.79073521

  Status: nONE
```

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state



Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 5-Body RDM (Sites 3,4,5,0,1)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3994863173
  Von Neumann entropy  : 1.1573895658
  Eigenvalue min       : -5.214677e-17
  Eigenvalue max       : 4.456381e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.39948632

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

STATUS: MIXED STATE
___________________________________

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

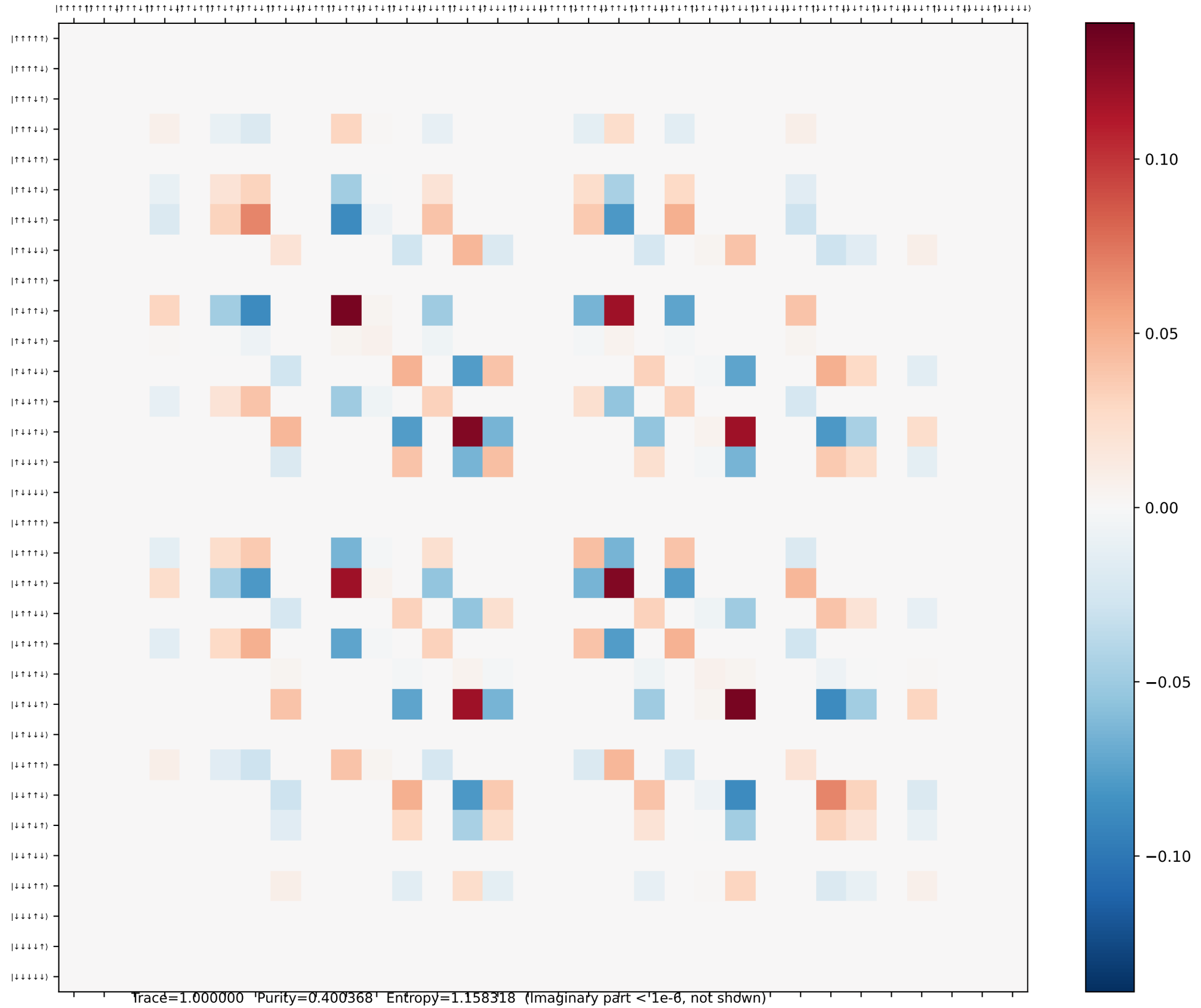
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.61290717   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.79129919

  Status: nONE
```

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

5-Body RDM (Sites 4,5,0,1,2)

Real Part



Trace=1.000000 Purity=0.400368 Entropy=1.158318 (Imaginary part <1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 5-Body RDM (Sites 4,5,0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4003682228
  Von Neumann entropy  : 1.1583182995
  Eigenvalue min       : -2.829789e-17
  Eigenvalue max       : 4.462601e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.40036822

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

STATUS: MIXED STATE
___________________________________

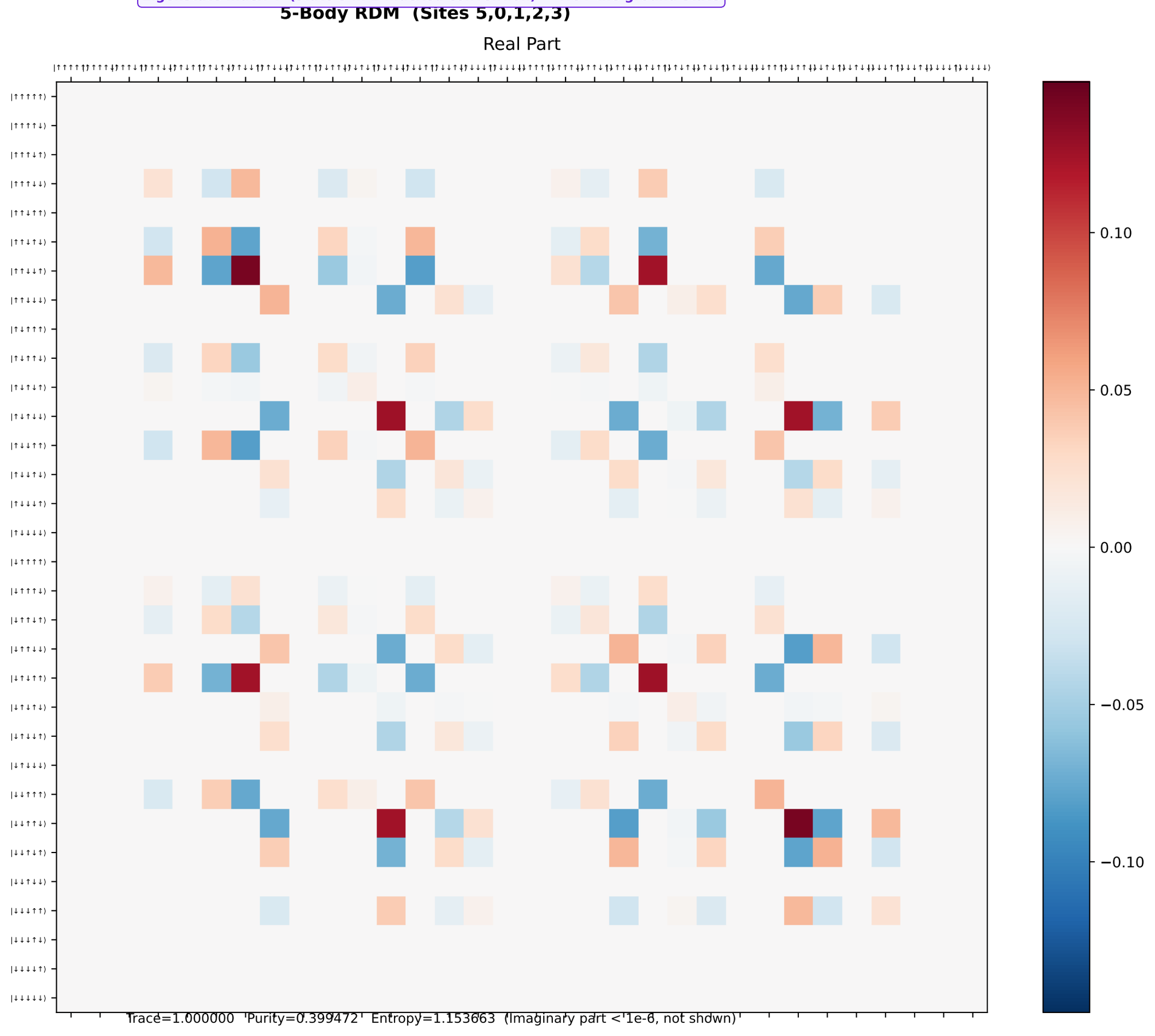
Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.63646762   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.92690926

  Status: nONE
```

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state



Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 5-Body RDM (Sites 5,0,1,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.3994720399
  Von Neumann entropy  : 1.1536633312
  Eigenvalue min       : -4.846925e-17
  Eigenvalue max       : 4.455320e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.39947204

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

STATUS: MIXED STATE
___________________________________

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.59519930   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.79090187

  Status: nONE
```